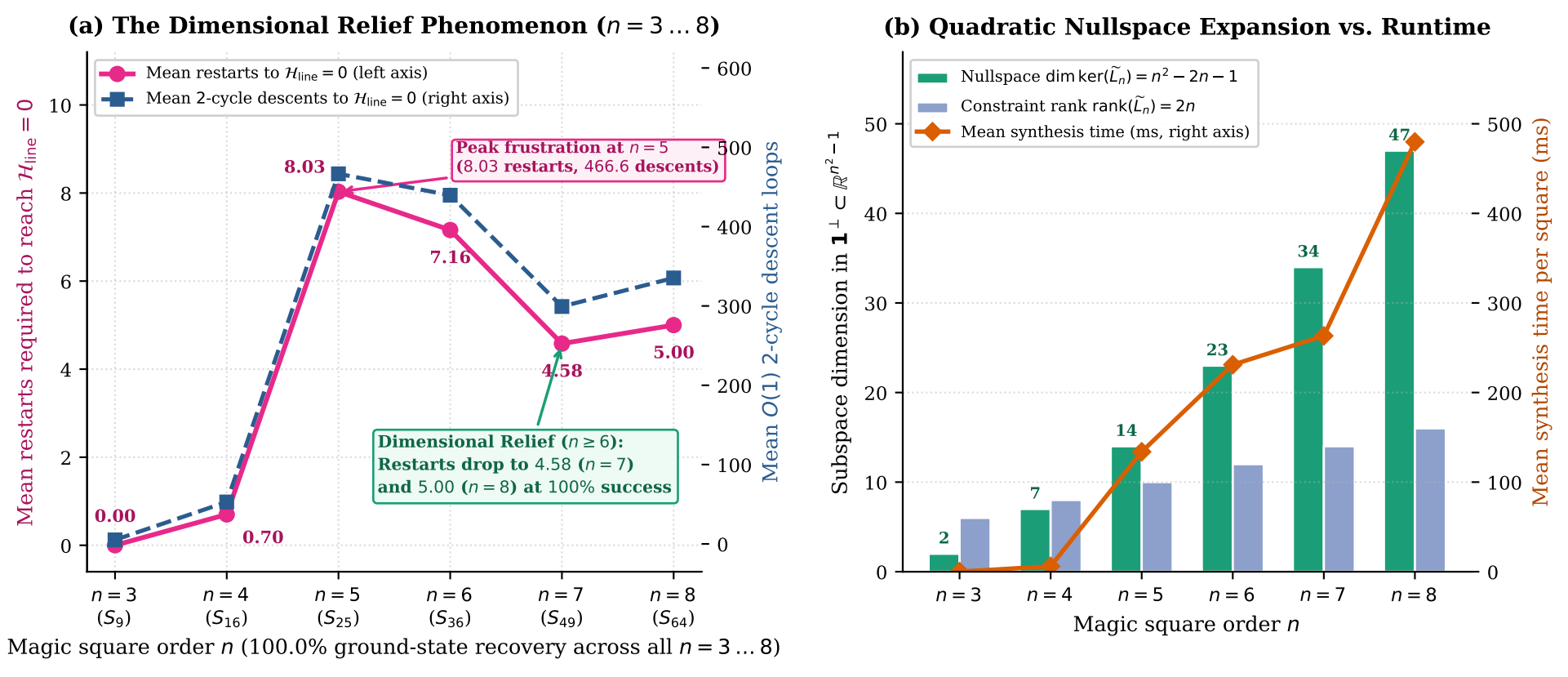

(a) The Dimensional Relief Phenomenon ($n = 3 \ldots 8$)
Mean restarts to $\mathcal{H}_{\text{line}} = 0$ (left axis)
Mean 2-cycle descents to $\mathcal{H}_{\text{line}} = 0$ (right axis)
Mean restarts required to reach $\mathcal{H}_{\text{line}} = 0$
Mean $O(1)$ 2-cycle descent loops
Peak frustration at $n = 5$
(8.03 restarts, 466.6 descents)
8.03
7.16
4.58
5.00
0.00
0.70
Dimensional Relief ($n \geq 6$):
Restarts drop to 4.58 ($n = 7$)
and 5.00 ($n = 8$) at 100% success
$n = 3$ (S_9)
$n = 4$ (S_{16})
$n = 5$ (S_{25})
$n = 6$ (S_{36})
$n = 7$ (S_{49})
$n = 8$ (S_{64})
Magic square order n (100.0% ground-state recovery across all $n = 3 \ldots 8$)
(b) Quadratic Nullspace Expansion vs. Runtime
Nullspace $\dim \ker(\widetilde{L}_n) = n^2 - 2n - 1$
Constraint rank $\operatorname{rank}(\widetilde{L}_n) = 2n$
Mean synthesis time (ms, right axis)
Subspace dimension in $\mathbf{1}^{\perp} \subset \mathbb{R}^{n^2-1}$
Mean synthesis time per square (ms)
2
7
14
23
34
47
Magic square order n